Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: *WXIMPACTBENCH* | ***Context:*** *Gran Chaco Tri-Border Emergency Response Zone*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's multi-label simultaneous binary classification output [Q57, Q117] aligns with the deployment's minimum acceptable output form. Dataset analysis confirms the schema produces compact six-binary label vectors per row [D3, D10], operationally usable for national coordinator dashboards. However, the evaluation framework has notable gaps: F1, accuracy, and row-wise accuracy [Q43, Q45] aggregate across categories and do not differentiate field-gendarme rapid-lookup needs from national-coordinator aggregate-situational-awareness needs. The ranking-based QA task — which the deployment user explicitly rejected as insufficiently reliable for direct operational use — is a substantial portion of the benchmark [Q3, Q15] and includes a known self-referential bias when GPT-4O is both question generator and ranker [Q88]. No confidence scores or probability distributions are produced, although the user indicated these would be useful for human-review-mediated workflows. Score is 3 (mixed) because the classification output structure is genuinely usable, but evaluation-framework relevance to the role-differentiated deployment is partial.

ID	Evidence
Q3	“The benchmark involves two evaluation tasks, multi-label classification and ranking-based question answering.” (p.1)
Q15	“With our constructed dataset, we develop a benchmark, WXIMPACTBENCH, to investigate the capacity of LLMs to understand disruptive weather impacts with two tasks: i) multi-label classification and ii) ranking-based question-answering.” (p.2)
Q43	“For multi-label classification task, we use F1-score, accuracy, and row-wise accuracy as evaluation metrics.” (p.6)
Q45	“Compared to the common F1-score and accuracy, the row-wise accuracy is a strict metric that requires more accurate output as the model should correctly classify all six impact labels for a given article.” (p.6)
Q57	“Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once.” (p.6)
Q88	“Notice that the ranking results would contain bias when the evaluated model is used for question generation (GPT-4O in our cases). This is a common phenomenon (Zhou et al., 2023) and needs to be avoided in benchmarking.” (p.9)
Q117	“The classification task is binary (true/false), requiring the model to identify whether the text explicitly mentions any of the defined impacts.” (p.16)
D3	by six o'clock the electric car services in all the important points west of Toronto had been completely paralyzed — 1894 blizzard article: infrastructure disruption to Canadian railway/electric car — exemplary on-target sample
D10	Residents of the locality were put to great inconvenience by the huge piles of snow now accumulated on the streets... Already $30,000 has been expended in removing it — 1887 Montreal snow removal municipal debate — multi-label (infra+pol+fin) correctly labeled

Strengths

- Multi-label simultaneous binary output [Q57] is the minimum acceptable form for the deployment
- Dataset confirms compact binary-vector schema works in practice [D3, D10]
- Standard, well-known classification metrics (F1, accuracy) [Q43] are interpretable by the deployment team
- Modest computational requirements [Q126, Q128] make local replication feasible for deployment teams

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Q43	“For multi-label classification task, we use F1-score, accuracy, and row-wise accuracy as evaluation metrics.” (p.6)

Q57	“Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once.” (p.6)
Q126	“For the large proprietary models (e.g., GPT-4o), conducting a one-time evaluation on our WXImpactBench costs approximately $3 for multi-label classification tasks and $5.5 for ranking-based QA tasks.” (p.17)
Q128	“The relatively modest computational requirements highlight the accessibility of our benchmark for researchers with limited computational resources, while still enabling comprehensive evaluation of state-of-the-art models” (p.17)
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (text-based binary multi-label) matches deployment need for simultaneous classification [Q57], but does not provide confidence scores requested for human-review-mediated workflows.

OF-2: Not applicable — speech-based output is not a deployment requirement.

OF-3: Not directly applicable to text-classification output. Literacy of operators is not in question; the user noted intermediate English proficiency for technical practitioners (system operates in Spanish) [regional english_proficiency].

OF-4: Partial external validity gap: F1 and row-wise accuracy [Q43, Q45] do not distinguish field-gendarme rapid-lookup precision needs from national-coordinator aggregate-recall needs; ranking task evaluation is not deployment-relevant per user [regional output_format_requirements].

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Q43	“For multi-label classification task, we use F1-score, accuracy, and row-wise accuracy as evaluation metrics.” (p.6)
Q45	“Compared to the common F1-score and accuracy, the row-wise accuracy is a strict metric that requires more accurate output as the model should correctly classify all six impact labels for a given article.” (p.6)
Q50	“For the ranking-based QA task, we deploy the standard metric that emphasizes the accuracy of top positions for evaluation, including Hit@1, nDCG@5, Recall@5, and MRR.” (p.6)
Q57	“Different from traditional methods that decompose multi-label text classification into multiple binary classification tasks (Boutell et al., 2004; Liu et al., 2017), we simultaneously identify all relevant disruptive weather impacts for each input by calling the LLM once.” (p.6)
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Information Gaps

- Whether benchmark-style F1 scores predict trust-in-output for field gendarmes vs. national coordinators is unmeasured; no role-differentiated evaluation exists

Requires Expert Verification

- AFE/SINAGIR coordinators should specify whether per-class precision floors or aggregate recall ceilings are operationally binding metrics

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Remediation

Gap 1: Aggregate F1/row-wise accuracy metrics do not distinguish field-gendarme precision needs from national-coordinator aggregate-recall needs; no confidence scores produced.

Recommendation: Define role-differentiated evaluation metrics: per-event precision floors for field gendarme rapid-lookup; aggregate per-category recall and calibrated multi-event coverage for national coordinators; expose model probability distributions to support human-in-the-loop review.